

Challenges Faced

- Spark Environment Setup: Setting up Apache Spark and configuring SPARK_HOME and the required environment variables correctly was initially challenging.
- Dataset File Path: The dataset filename contained spaces (movie ratings.csv), so the correct filename had to be used while loading the CSV file.
- Schema and Data Loading: Loading the CSV with header=True and inferSchema=True and ensuring that columns such as movieId and rating were correctly identified required verification.
- Partition Management: Understanding the initial number of partitions and correctly applying repartition() with different partition counts required testing.
- Performance Measurement: Measuring execution time accurately using time.perf_counter() was challenging because execution time can vary between Spark runs.
- Caching and Persistence: Understanding the difference between cache() and persist(StorageLevel.DISK_ONLY) and triggering caching using an action such as count() required practical experimentation.
- Comparing Optimization Techniques: Comparing caching, repartitioning, and coalescing was challenging because their performance depends on dataset size, workload, and partition configuration.